

GIT 3: Quiver Representations

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1 Problem

In the previous talks, Jonathon and Daniel set up the general framework of trying to parameterize something using GIT. In this talk, we'll focus on a specific moduli problem which we can solve pretty explicitly with GIT. Before getting into this problem, I wanted to at least mention some famous applications:

- Moduli of projective hypersurfaces
- Moduli of vector bundles on a curve
- Moduli of pure sheaves (i.e, having pure dimensional support) on a general variety (due to Gieseker, Simpson, Maruyama, etc).
- Moduli of curves of stable curves.

Today I wanted to focus on an application that is in some sense more modest, but still leads to very interesting geometry.

1.1 Quiver Representations

Definition 1.1. A **quiver** is a directed graph Q with set of vertices denoted Q_0 and set of arrows denoted Q_1 , where we specify an arrow by source and target, which assembles into functions

$$s, t : Q_1 \rightarrow Q_0.$$

Example 1.2. The **Kronecker quiver** is the directed graph



which we could write combinatorially as $Q_0 = \{0, 1\}$, $Q_1 = \{\alpha, \beta\}$, and $s(\alpha) = s(\beta) = 0$, $t(\alpha) = t(\beta) = 1$.

Definition 1.3. A **quiver representation** (over k) (M, φ) is an assignment of vertices i to k -modules M_i and arrows to k -module homomorphisms $\varphi_\alpha : M_{s(\alpha)} \rightarrow M_{t(\beta)}$.

The **dimension vector** of a quiver representation is the tuple of ranks of the M_i , i.e., the tuple $(\dim_k M_i)_{i \in Q_0}$.

A **morphism** $f : M \rightarrow N$ of quiver representations is the data of k -module homomorphisms $f_i : M_i \rightarrow N_i$ commuting with the arrows (like a chain complex morphism).

Example 1.4. Consider the quiver

$$\bullet \xrightarrow{\alpha} \bullet$$

Fixing a dimension vector (d_0, d_1) , what are *all* of the quiver representations over $\mathbb{C}$ (up to isomorphism)?

The data of a $\varphi_\alpha : M_0 \rightarrow M_1$ is just a matrix $\mathbb{C}^{d_0} \rightarrow \mathbb{C}^{d_1}$, and the equivalence relation of “up to isomorphism” allows us to change bases on both sides independently. By Gaussian elimination, we can change basis to a matrix with a block which is the identity I and 0s everywhere else, so the φ_α is determined just by fixing a rank between 0 and $\min\{d_0, d_1\}$.

So, there is no moduli for these representations—this is just a discrete natural number.

Example 1.5. However, once we move up to the Kronecker quiver, once can convince oneself that representations are the same thing as complexes

$$\mathcal{O}_{\mathbb{P}^1}(-1)^{d_0} \xrightarrow{\varphi} \mathcal{O}_{\mathbb{P}^1}^{d_1}$$

up to isomorphism. For the case $d_0 = d_1 = 1$, such a complex is determined by the cokernel, which is either $\mathcal{O}_{\mathbb{P}^1}$ if $\varphi = 0$, or a skyscraper sheaf $\mathcal{O}_P$. This suggests that (throwing away the exceptional case of $\varphi = 0$), we get a bijection between representations and $\mathbb{P}_{\mathbb{C}}^1$.

1.2 Families and Moduli

We will formalize this as a moduli problem.

Problem 1.6. A **family** of quiver representations of dimension vector $\underline{d}$ over a scheme T/k is the data of

- a collection of locally free sheaves $\mathcal{F}_v$ for $v \in Q_0$ of rank d_v .
- a collection of sheaf morphisms $\varphi_\alpha : \mathcal{F}_{s(\alpha)} \rightarrow \mathcal{F}_{t(\alpha)}$.

There is then an equivalence relation on these families, where $(\mathcal{F}_v, \varphi_\alpha) \sim (\mathcal{G}_v, \psi_\alpha)$ if there exist isomorphism $\mathcal{F}_v \cong \mathcal{G}_v$ commuting with the $\varphi_\alpha, \psi_\alpha$.

This gives us a moduli problem—we now can say what a family of quiver representations is. We can ask whether the following functor has a fine/coarse moduli space:

$$\begin{aligned} (\text{Sch}/k)^{\text{op}} &\longrightarrow \text{Set} \\ T &\longmapsto \{\text{families over } T\} / \sim . \end{aligned}$$

This one will generally not—instead we’re going to have to restrict ourselves to “(semi)stable” quiver representations, and identify S -equivalent semistable quiver representations.

2 Stability

Let's arrive at the notion of the stability of a quiver representation by following the GIT story. If I want to parameterize quiver representations of Q with dimension vector $\underline{d}$, this is what I should do:

1. (Redundantly) parameterize quiver representations by writing down all possible matrices for the arrows, i.e., write down the variety

$$X = \prod_{\alpha \in Q_1} \mathbb{A}_k^{d_s(\alpha) \cdot d_t(\alpha)}$$

which has k -points $(X_\alpha)_{\alpha \in Q_1}$ matrices.

2. Remove the redundancy by quotienting out by change of basis. Define the “change of basis” group

$$G = \left(\prod_{v \in Q_0} \mathrm{GL}_{d_v}(k) \right) / \mathbb{G}_m$$

(where we mod out by the extra copy the $\mathbb{G}_m$ in the center of the action, i.e., scaling all the matrices at once), where a k -point $(P_v)_{v \in Q_0}$ acts on a k -point $(X_\alpha)_{\alpha \in Q_1}$ by

$$(P_v)_{v \in Q_0} \cdot (X_\alpha)_{\alpha \in Q_1} = (P_{t(\alpha)} X_\alpha P_{s(\alpha)}^{-1}).$$

Note, we quotient by $\mathbb{G}_m$ so that maximal dimensional orbits will have any hope of having dimension $\dim G$ (but as Daniel mentioned, we could just replace this notion of having $\dim G$ orbits/0-dimensional stabilizers with the notion of maximal dimension to define notions of stability).

Now, we want to come up with an interesting quotient $X//G$.

2.1 Approach 1: Affine GIT

We could try to do affine GIT, since we are quotienting an affine space anyways. However, when our quiver is connected and has no any directed cycles, this ends up being $\mathrm{Spec} k$, since

$$X \xrightarrow{\pi} X//G$$

is a good quotient, so it is in particular separated and is constant on k -orbits, and so for any assignment of matrices $x \in X$, it sends

$$\pi(\overline{G \times x}) \subseteq \overline{\pi(G \times x)},$$

but we claim that the 0 assignment is dense in *every* orbit, so actually

$$\pi(x) \in \pi(\overline{G \times 0}) \subseteq \overline{\pi(G \times 0)} = \{\pi(G \times 0)\}.$$

Then, to see that 0 is dense in every orbit, we observe the 1-parameter subgroup for $t \in \mathbb{G}_m$

$$t \mapsto (t^{\pm 1} I_{d(v)})_{v \in Q_0} \in G$$

(using the no directed cycles to alternate the signs) which acts on matrix assignments by (without loss of generality for parity)

$$t \cdot (A_v) = (tIA_v(t^{-1}I)^{-1}) = (t^2A_v),$$

so the G orbit of A_v has a closure which hits 0 as $t \rightarrow 0$.